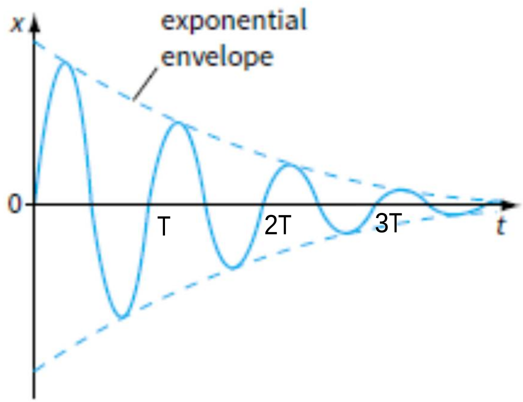
the skateboarder exits at point B will be higher.

3(a)(i) As the block oscillates in the liquid, energy is needed to overcome the resistive (viscous) forces present in the liquid. Energy of the oscillating system thus decreases, resulting in damped oscillations.

(ii)



(b)(i) The motion of the block is in **forced oscillation** as the water waves provides the external periodic driving force to continuously supply energy to the block to compensate for the loss of energy due to damping. The block will therefore oscillate with a constant amplitude and at the frequency of the water waves.

When the driving frequency of the water waves matches the natural frequency of the block in the water, **resonance** occurs. At this particular frequency, there is maximum transfer of energy from the water waves to the block, so the block oscillates with the maximum amplitude.

(ii) 1. $$v = f\lambda \Rightarrow f = \frac{v}{\lambda} = \frac{0.90}{0.30} = 3.0\,\mathrm{Hz}$$

2. $$f = \frac{1}{2\pi}\sqrt{\frac{28}{m}} \Rightarrow m = \frac{28}{4\pi^2 f^2} = \frac{28}{4\pi^2(3.0)^2} = 78.8\,\mathrm{g}$$

3. When the block has absorbed some water, its mass increases, so its natural frequency becomes lower, so resonance no longer occurs. Hence the amplitude of vertical oscillation will be lower.

4(a) The increase in internal energy of a system is equal to the sum of the heat supplied